TORQUE LAB: INPUT CONDITIONS LIST WORKSHEET

COMPLETE THE ADDRESS AND INSTRUCTION.
Instructions with double asterisks ** are parallel branches.

RUNG 0: DOWN SOLENOID

XIC START_NO_ 1 START 1 pushbutton.

XIC START_NO_ 2 START 2 pushbutton.

_____ ____________________ Rung will go false when the timeout preset has been reached. What is the Tag of the status bit for the timeout?

What is the instruction type which will allow the down solenoid to be energized before the timeout?

[ ] ** TORQUE_1_SET Tag for the internal that are set when TORQUE 1 is reached.

[ ] ** TORQUE_2_SET Tag for the internal that are set when TORQUE 2 is reached.

[ ] ** TORQUE_3_SET Tag for the internal that are set when TORQUE 3 is reached.

[ ] ** TORQUE_4_SET Tag for the internal that are set when TORQUE 4 is reached.

For the above TORQUE 1 thru TORQUE 4, what instruction type will allow the down solenoid to be energized before the torque is reached?

RUNG 1: NUT RUNNER MOTORS: MOTOR_1, MOTOR_2**, MOTOR_3**, MOTOR_4** (Multiple output branches)**

Before the output branches:

_____ DN_LS An input which senses the nut runners are lowered.

_____ TO.DWELL.DN Time out status bit.

Each motor is placed on a parallel output branch. What internal Tag is used to turn off the individual in the motor output branch?

______	Torque_1_Set	Internal that turns off MOTOR 1.
______	Torque_2_Set	Internal that turns off MOTOR 2.
______	Torque_3_Set	Internal that turns off MOTOR 3.
______	Torque_4_Set	Internal that turns off MOTOR 4.

For MOTOR 1, MOTOR 2, MOTOR 3, and MOTOR 4 what instruction type will allow the motor to be energized before the torque is reached?

RUNG 2: **TIME OUT (RTO Timer)**

[ ]	DN_LS	An input which senses the nut runners are lowered.

RUNG 3: **TIME OUT ERROR (Latch Output)**

[ ]	TO_DWELL_DN	What status bit Tag will be set when the time for tightening the nuts has been exceeded?

What type of output will remain ON if the rung goes false?

RUNG 4: **ERROR RESET** (Unlatch Output) and TIME OUT RESET** (Reset Instruction), (Multiple output branches)**

[ ]	START_ NO_1	START 1 pushbutton have been released.
[ ]	START_ NO_2	START 2 pushbutton have been released.

What instruction type will be true when the button is released?

RUNG 5: TORQUE 1 SET

What Tag contains the scaled value from the analog torque sensor for Motor 1?

__________ ' AIN_1 ________________ A comparative instruction that will turn on the output when the torque value is within the limit.

RUNG 6: TORQUE 2 SET

What Tag contains the scaled value from the analog torque sensor for Motor 2?

__________ AIN_2 ________________ A comparative that will turn on the output when the torque value is within the limit.

RUNG 7: TORQUE 3 SET

What Tag contains the scaled value from the analog torque sensor for Motor 3?

__________ AIN_3 ________________ A comparative that will turn on the output when the torque value is within the limit.

RUNG 8: TORQUE 4 SET

What Tag contains the scaled value from the analog torque sensor for Motor 4?

__________ AIN_4 ________________ A comparative that will turn on the output when the torque value is within the limit.